

Fundamental Issues and Technical Problems About Pulsating Heat Pipes

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Since their introduction in the early 1990s, pulsating heat pipes (PHPs) have received a lot of attention due to their obvious advantages such as their geometrical simplicity, and their potential for high-heat flux applications even without power consumption. Although numerous investigators have studied PHPs over the last three decades, there still exist a few controversial issues on fundamental characteristics and several technical problems in practical applications. To put the finishing touches to the controversial issues and to shed light on the technical problems, recent advances in PHPs are critically reviewed in this paper. The results of this critical review are classified into two categories: (i) fundamental aspects of PHPs and (ii) practical aspects of PHPs. The first category focuses on reviewing the current state-of-the-art fundamental characteristics of PHPs. The second category summarizes the technical problems that are resolved for utilizing PHPs in practical applications. This review paper would help researchers or engineers who are working on or utilizing PHPs. [DOI: 10.1115/1.4051465]

1 Introduction

As the performance of electronic devices increases, thermal management has a significant role in accelerating performance enhancements and extending the lifetime of electronics devices. Thus, many cooling solutions have been proposed, including a pulsating heat pipe (PHP), which was first introduced by Akachi et al. [1]. Among the various types of heat pipes, PHPs have received a lot of attention as an efficient cooling device due to their distinct advantages such as a simple wickless design and high maximum allowable heat flux. For these reasons, the number of papers on PHPs has escalated recently [2]. Despite the fact that more than 1300 papers on PHPs have been published over the last decade, a complete understanding of the physical phenomena inside PHPs has not yet been achieved mainly due to the inherent complexity of thermohydrodynamic coupling inside PHPs.

To provide explanations on the physical phenomena inside PHPs, many review articles have been presented [3–10]. In 2008, Zhang and Faghri [3] reviewed the associated advances and unsolved issues. Their review paper was the first one on PHPs and provided insights into the fundamental characteristics of PHPs. As attempts to understand these fundamental characteristics of evolved, other review papers followed to report the local hydrodynamics of the oscillating flow [4], the effect of the thermophysical properties [5], parametric studies on design variables [6–8], and novel experimental techniques [6,8]. Ten years after the first review paper on PHPs was reported, Marengo and Nikolayev [9,10] provided a comprehensive overview of results from almost 200 papers. They compared experimental results with numerical results to bridge the gap between experimental and numerical observations. As they pointed out, a technology readiness level of PHPs is lower than 5, which is a minimum level for utilizing PHPs in practical applications. To utilize PHPs in practical applications, some questions, such as how to optimize channel geometries, how to select a working fluid, and how to fabricate flexible PHPs, should be answered. In addition, it should be noted that there still exist some misunderstandings about fundamental characteristics of PHPs and they need to be corrected.

The purpose of this review paper on PHPs is to put the finishing touches to some controversial issues on fundamental characteristics and to shed light on technical problems in practical applications. To this end, advances in PHPs over the recent decade are critically reviewed. Our group's research outcomes supported by a creative research initiatives grant for ten years as well as the results of other investigations are presented in this paper. The contents of this paper are classified into two categories: (i) fundamental aspects of PHPs and (ii) practical aspects of PHPs. The first category is related to the controversial issues that still exist among those who are working on PHPs. These include (a) operating mechanisms of PHPs, (b) unstable operation of PHPs, and (c) numerical models for PHPs. The second category presents several technical problems to be resolved for practical applications. The technical problems that need to be overcome are as follows: (a) omnidirectional performance, (b) extension of the operating range, (c) operation under the local heating condition, (d) optimization of the channel diameter and the number of turns, (e) comparison between closed-loop and closed-end PHPs, (f) selection of the working fluid, and (g) fabrication of flexible PHPs.

2 Fundamental Aspects of Pulsating Heat Pipes

2.1 Operating Mechanisms of Pulsating Heat Pipes

2.1.1 Heat Transfer Mechanisms of PHPs. In the last decade or so, there has been a controversy on the proportions of sensible and latent heat to the overall heat transferred through PHPs. Some researchers [3,6,11–13] have claimed that heat transfer in the form of sensible heat is a dominant heat transfer mechanism of PHPs. Senjaya and Inoue [13] have developed a numerical model to obtain the proportion of sensible heat to the overall heat. Based on their model, the ratio of sensible heat to overall heat is 74%. However, conflicting results regarding the contributions of sensible/latent heat have been also reported [14,15]. Nikolayev [14] has claimed that the contribution of latent heat is considerable based on his numerical model for PHPs. In his numerical results, the contribution of latent heat is 82% in the evaporator section and 71% in the condenser section. To conclusively determine the contributions of sensible and latent heat, Jo et al. [16] have conducted experimental studies to measure the temperature and heat flux distributions of PHPs. For this, IR thermometry, which was previously applied to measure the temperature and

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heat flux distributions on a boiling surface [17], is applied to measure the distributions of the temperature and heat flux of PHPs with high spatial and temporal resolutions. In addition, flow visualization is synchronized with temperature and heat flux distributions to identify the local flow patterns corresponding to the local temperature and heat flux distributions. As shown in Fig. 1, time-resolved distributions of heat flux and temperature along with flow visualization are obtained at several successive frames. Using the experimental data, the contribution of latent heat is estimated to be between 66% and 74%. This indicates that heat transfer in the form of latent heat is a dominant heat transfer mechanism of PHPs. Additionally, since PHPs transfer heat through fluid motion, the effect of a flow behavior on heat transfer inside PHPs is investigated. Jung and Kim [18] quantitatively measured the proportion of latent heat to overall heat with an increasing oscillation amplitude based on the IR thermometry. As the input power increases, the proportion of latent heat continues to increase with the increasing oscillation amplitude. Because heat transfer in the form of latent heat is a dominant heat transfer mechanism of PHPs, an increased proportion of latent heat leads to an improvement of the thermal performance of the PHP. Therefore, it can be

concluded that the thermal performance of PHPs can be increased by increasing either the amplitude or the proportion of latent heat to overall heat transferred through PHPs.

2.1.2 Flow Behavior and the Thermal Performance of PHPs.

Due to the complex flow behavior inside PHPs, flow visualization technique has been useful in understanding operating mechanisms of PHPs. To visualize the fluid flow inside PHPs, many researchers fabricated a whole section of PHPs or part of it using transparent materials. Khandekar and Groll [19] have reported that the flow behavior inside PHPs changes from an oscillation mode to a circulation mode, which in turn increases the thermal performance of PHPs. Liu et al. [20] have conducted flow visualization experiment using a high-speed camera. They have shown that as the channel diameter increases, the flow behavior inside PHPs changes from an oscillation mode to a circulation mode with increasing thermal performance of PHPs. Some researchers have fabricated flat plate PHPs covered by glass to visualize the entire section of PHPs [21–29]. Based on the flow visualization results of the whole part of PHPs, it is found that the circulating flow inside PHPs assists vapor plugs to penetrate deeply into the

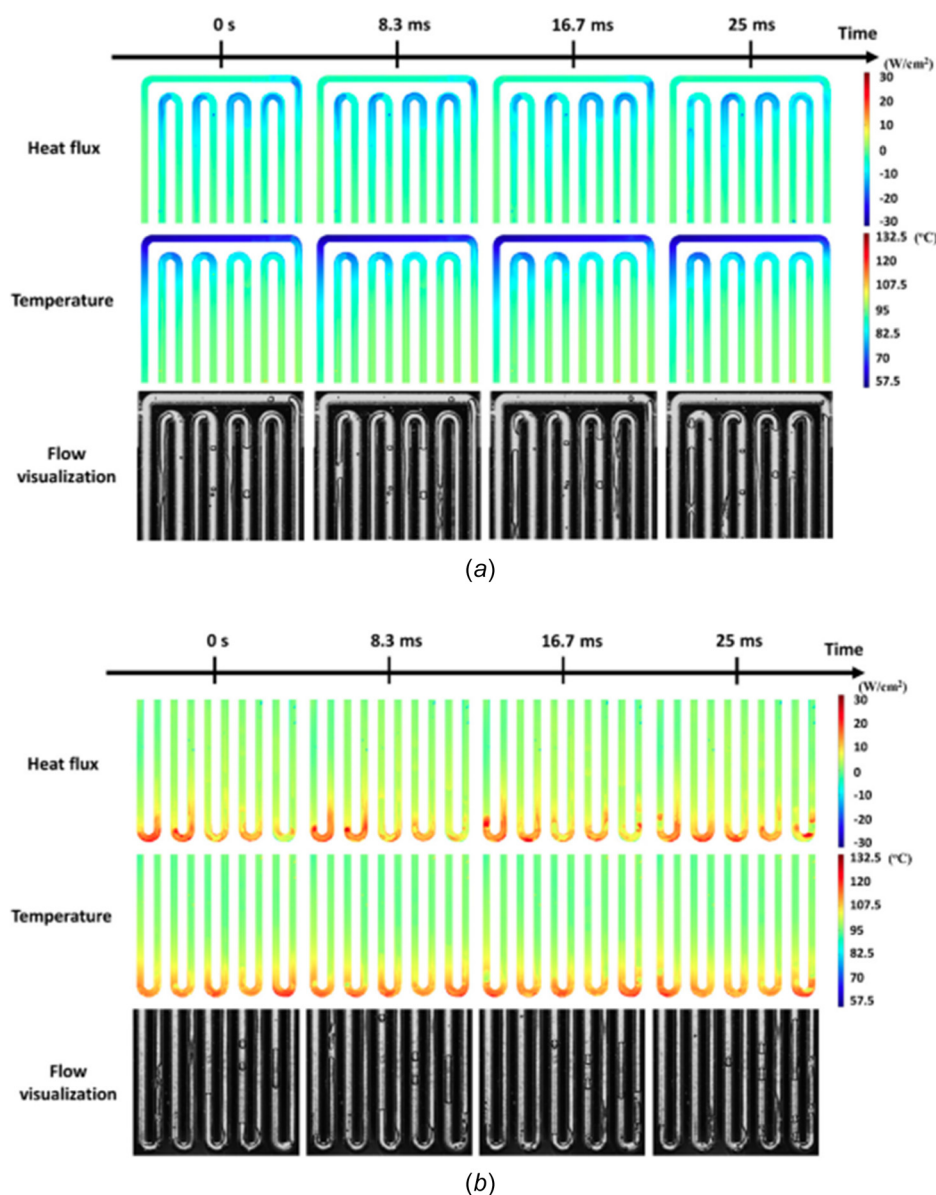


Fig. 1 Time-resolved distributions of heat flux and temperature with the corresponding flow visualization: (a) at the condenser section and (b) at the evaporator section

condenser section [21,26,29]. Similarly, an increase of the amplitude of oscillating liquid slugs leads to a deep penetration of vapor plugs into the condenser section [24,25,27]. These changes result in an improvement of the thermal performance of PHPs. To quantitatively explain how the flow behavior inside PHPs affects the thermal performance of PHPs, an average volumetric fraction in the condenser section is introduced [26,29]. The average volumetric fraction in the condenser section, $\bar{\beta}_c$, is defined as follows:

$$\bar{\beta}_c \equiv \frac{\bar{A}_{vp,c}}{A_c} = \frac{\bar{A}_{2\phi,c}}{A_c} = \frac{\sum_{i=1}^{\infty} \bar{L}_{2\phi,i}}{L_c} \quad (1)$$

where $\bar{A}_{vp,c}$, A_c , $\bar{A}_{2\phi,c}$, $\bar{L}_{2\phi}$, and L_c are the time-averaged surface area occupied by the vapor plugs enclosed with liquid films in the condenser section, the total surface area of the channels in the condenser section, the time-averaged surface area where phase change heat transfer occurs in the condenser section, the time-averaged length of the vapor plug enclosed with liquid film and the length of the condenser section, respectively. It is experimentally found that the average volumetric fraction in the condenser section is a major contributor to the thermal performance of PHPs [26,29]: The thermal performance of PHPs is improved as the average volumetric fraction in the condenser section is increased. Based on the fact that the average volumetric fraction in the condenser section is physically related to the surface area where heat is transferred in the form of latent heat, it can be inferred that the proportion of latent heat to overall heat is increased as the average volumetric fraction in the condenser section increases. Therefore, an increase of the average volumetric fraction in the condenser section enhances the thermal performance of a PHP due to an increased proportion of latent heat over all heat transferred through PHPs.

2.2 Unstable Operation. Unstable operations of PHPs lead to a noticeable deterioration in the thermal performance of PHPs. Among various unstable operations of PHPs, dryout and stopover are the most representative phenomena resulting in a significant temperature increase in PHPs. Section 2.2.1 covers a dryout phenomenon, and Sec. 2.2.2 covers a stopover phenomenon.

2.2.1 Dryout phenomenon. Previous studies have reported that as the input heat flux increases above the maximum allowable heat flux, it is observed that the operating limit of PHPs occurs with a dryout phenomenon in the evaporator section [30–33]. Hence, many researchers have investigated the operating limit of PHPs to ascertain their operating range. Kammuang-Lue et al. [31] have conducted experiments on glass-based PHPs to visualize the dryout phenomenon in the evaporator section. Based on the experimental data, they have reported an empirical correlation for predicting the maximum allowable heat flux of PHPs. However, their correlation is limited to PHPs operating in a circulation mode. In general, PHPs are observed to operate either in a circulation mode or an oscillation mode. Moreover, some researchers have reported that a circulation mode is not observed in their PHPs [34]. Hence, to expand the applicable range of the correlation for predicting the maximum allowable heat flux of PHPs, Lee and Kim [35] and Lee et al. [36] have conducted a series of experiments on silicon-based PHPs to suggest a new correlation for predicting the operating limit of PHPs in an oscillation mode. It is experimentally found that the liquid supply via the motions of liquid slugs is a key factor that explains dryout mechanisms in PHPs. When the rate of the liquid supply is smaller than the evaporation rate, dryout occurs. The maximum allowable heat flux for PHPs in a vertical orientation is proportional to the third power of the channel diameter regardless of the cross-sectional shape. Using the developed correlation, a contour map of the maximum allowable heat flux with respect to the number of turns and the inclination angle is presented in Fig. 2.

2.2.2 Stopover Phenomenon. According to the previous studies, PHPs show degradation in the thermal performance with a

stopover phenomenon or do not operate at all in a horizontal orientation unless they have a large number of turns (typically more than 20 turns) [3,14,24]. The stopover phenomenon, which was first reported by Khandekar and Groll [37], is defined as a repeated very small-amplitude oscillation between intermittent large-amplitude oscillations. In the stopover period, the liquid slugs do not penetrate into the evaporator section, which causes a significant increase in the evaporator temperature [38,39]. Hence, the stopover phenomenon causes the performance degradation of PHPs in a horizontal orientation. To prevent stopover from occurring in PHPs, Jun and Kim [40] have developed a criterion for stable operation of PHPs even in a horizontal orientation. They have treated PHPs as mass-spring-damper systems, and have introduced an effective damping coefficient, which is the sum of a damping coefficient for a liquid slug and damping coefficients for two neighboring vapor plugs. Also, they have experimentally shown that the transition from (to) stopover to (from) a large-amplitude oscillation occurs when an effective damping coefficient becomes negative (positive) as depicted in Fig. 3. For PHPs filled with ethanol, the effective damping coefficient reduces to the ratio of the channel diameter to the effective length of PHPs (D_h/L_{eff}), and it has to be greater than approximately 0.03 for stable operation without any stopovers in a horizontal orientation.

2.3 Numerical Models for Pulsating Heat Pipes

2.3.1 Current status of Numerical Models. Due to the complex nature of thermohydrodynamic coupling inside PHPs, a numerical simulation of PHPs is a cost-effective method for predicting the thermal performance of PHPs. The numerical models can be classified into three generations depending on the degree of complexity in film dynamics as shown in Fig. 4. A common characteristic of the first-generation models [11,41–44] is an assumption that the thickness of the liquid film does not vary with space and time. Second-generation models [14,45] are based on the assumption that the length of the liquid film is allowed to be varied while the thickness of the liquid film is uniform. Nikolayev and Marengo [10] have summarized the pros and cons of the first and second-generation models. Hence, it is highly recommended to refer to their review paper for the details on the first- and second-generation models. This paper mainly focuses on third-generation models. The first- and second-generation models are somewhat limited because the thickness of the liquid film is predetermined and treated as a fitting parameter. To remove the fitting parameter with regard to the thickness of the liquid film, third-generation models [46,47] have been developed by allowing both the thickness and the length of liquid film to be varied temporally and spatially. The third-generation model developed by Bae

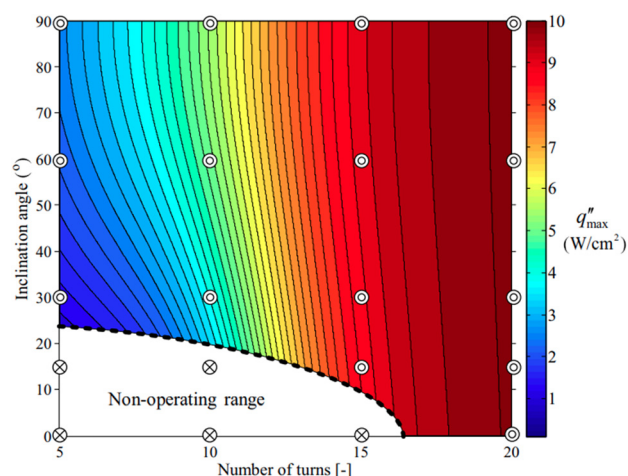


Fig. 2 Contour map of the maximum allowable heat fluxes in terms of the number of turns and the inclination angle

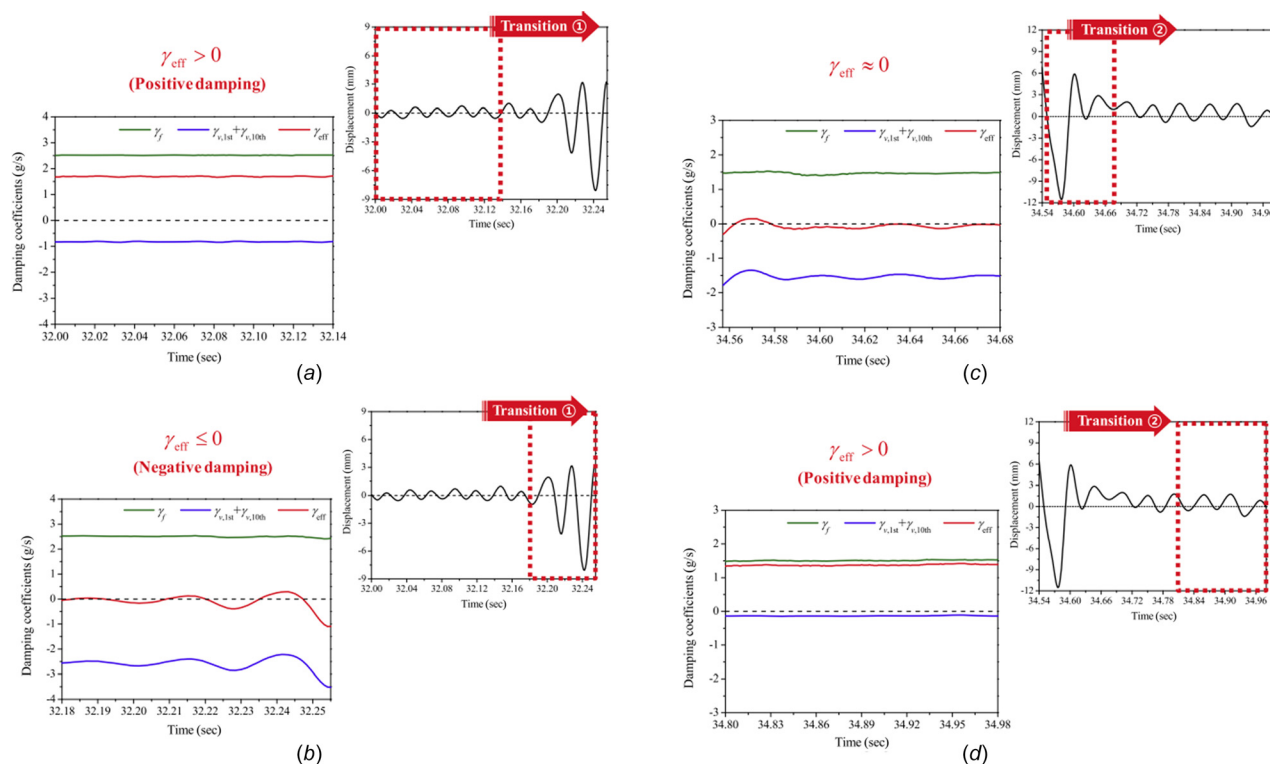


Fig. 3 Temporal variations of the damping coefficients: (a) in the stopover period before a large-amplitude oscillation, (b) during the transition to a large-amplitude oscillation, (c) during the transition to a stopover, and (d) in the stopover period

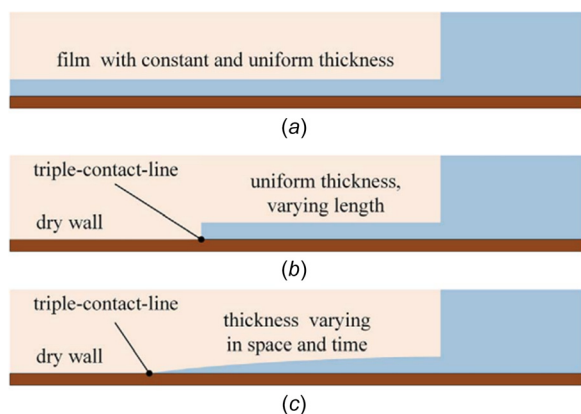


Fig. 4 Three generations of numerical models for predicting the thermal performance of PHPs: (a) first generation, (b) second generation, and (c) third generation

et al. [47] has been shown to be accurate in predicting experimental data within $\pm 20\%$ for various parameters, including number of turns, working fluid, filling ratio, and operating temperature; however, the third-generation model needs to be improved as far as the computational time and cost are concerned. For this, Noh and Kim [48] have developed a numerical model for PHPs with a lumped parameter approach in which each of the liquid slug and the vapor plug is regarded as a lumped mass.

The developed numerical models are able to predict the thermal performance of PHPs reasonably well during their stable operation. However, they show very limited success in predicting the unstable operation of PHPs, such as dryout. To the best of the authors' knowledge, a permanent dryout (device failure) has not been predicted by the numerical models yet. Even when the temporal and local dryout of a liquid film occurs in the numerical

simulation, the dry wall is rewetted immediately by adjacent liquid slugs, and a permanent dryout has not been numerically observed [47]. The physical reason for a permanent dryout is still unclear at the current state of understanding about PHPs. After revealing dryout mechanisms of PHPs, a numerical model for predicting not only the thermal performance of PHPs but also the operating range of PHPs should be developed.

2.3.2 Thermodynamic State of Vapor Plugs. To improve a numerical model for PHPs in terms of accuracy, it is essential to identify the thermodynamic state of vapor plugs because using an inappropriate equation of state for vapor plugs leads to errors in calculating the heat transfer rate through the vapor plugs. Some researchers have claimed that vapor plugs in PHPs are always saturated [42,49,50], while others have claimed that vapor plugs in PHPs are superheated [14,51]. To identify the thermodynamic state of vapor plugs in PHPs, experiments are performed in a single straight tube [49,52] or a two-turn PHP [53]. Nevertheless, there still exists controversy about the thermodynamic state of vapor plugs based on their experimental results. For this reason, Jun and Kim [54] have conducted experiments with a multiturn PHP to clarify the controversy about the thermodynamic state of vapor plugs in PHPs. It is confirmed that a vapor plug is saturated if the vapor plug is surrounded by a liquid film, while a vapor plug is superheated if the vapor plug is in direct contact with the dry wall without a liquid film. Controversial results in the previous studies have been clarified by using the above conclusion.

3 Practical Aspects of Pulsating Heat Pipes

3.1 Omnidirectional Performance. As mentioned in Sec. 2.2.2, PHPs with only a few turns show worse thermal performance or do not operate at all in a horizontal orientation. Many researchers have focused on flow restricting structures to ensure PHPs operate with only a few turns in a horizontal orientation. Pastukhov and Maydanik [55] have employed ball check valves for PHPs to

improve the thermal performance of PHPs in a horizontal orientation. Feng et al. [56] have integrated spring-loaded check valves on PHPs to decrease the orientation sensitivity of PHPs. Instead of using check valves, Chien et al. [57] have developed a PHP with a dual-diameter channel to provide a clue for achieving the omnidirectional performance even with a few turns. However, their PHP with a dual-diameter channel has at least 30% lower thermal performance in a horizontal orientation than in a vertical orientation. For this reason, Kwon and Kim [58] have optimized the difference in the channel diameter of a dual-diameter channel to achieve the omnidirectional performance even with a few turns. The optimum range of the difference in the channel diameter of a dual-diameter is experimentally investigated as shown in Fig. 5. It is experimentally confirmed that the omnidirectional performance occurs when the capillary pressure generated by the difference in the channel diameter is larger than the viscous pressure drop.

3.2 Extension of the Operating Range. In general, PHPs enter stable operation after startup. Therefore, if early startup is promoted, the operating range of PHPs can be extended. For this, many researchers [13,59,60] have claimed that startup can be improved by fabricating nucleation sites inside PHPs. For this reason, some investigators have used nanofluids in PHPs to fabricate nucleation sites for promoting nucleation, which leads to early startup [61–64]. As an alternative to fabricating nucleation sites inside PHPs, Kim and Kim [26] have employed artificial cavities in PHPs using MEMS techniques. As shown in Fig. 6, artificial

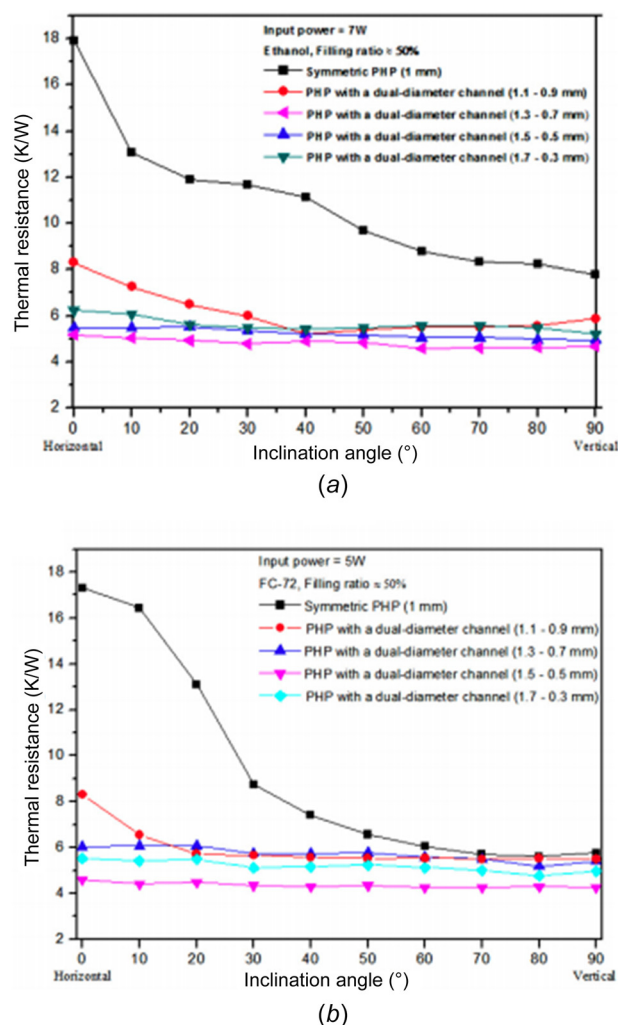


Fig. 5 The optimum range of the difference in channel diameter of a dual-diameter

cavities promote nucleation in the evaporator section. This leads to early startup and enhanced thermal performance of PHPs. Moreover, PHPs with each PHP having a different size of the artificial cavities are fabricated to investigate the effect of the size of artificial cavities on the thermal performance of PHPs. It is experimentally shown that a larger size of artificial cavities is helpful in promoting early startup, while it is unfavorable for enhancing the thermal performance of PHPs. By having artificial cavities of various sizes in a single PHP, both earlier startup and enhanced thermal performance of PHPs are achieved simultaneously.

3.3 Operation Under the Local Heating Condition. A PHP has a shortcoming when it is subjected to the local heating condition in which the size of the heating area is much smaller than a PHP. Thompson and Ma [65] have reported that under the local heating condition, nonoscillating liquid slugs are observed in some channels located away from channel. Nonoscillating liquid slugs deteriorate the thermal performance of PHPs. As the heating area decreases, the thermal performance and the operating range of PHPs are reduced simultaneously. Lim and Kim [28] have reported that rearranging the channels is one of the effective methods to prevent thermally isolated channels. They introduced an effective dissimilarity for designing a channel layout of a PHP under the local heating condition as depicted in Fig. 7(b). The effective dissimilarity, d_{eff} , is defined as the ratio of two dissimilarities as follows:

$$d_{\text{eff}} = \frac{d_1}{d_2} \quad (2)$$

$$d(A, B) = \left[\int_0^1 \{ \theta_A(s) - \theta_B(s) \}^2 dr \right]^{1/2} \quad (3)$$

where d_1 , d_2 , $d(A, B)$, θ_A , θ_B , and r are dissimilarity 1, dissimilarity 2, a dissimilarity between the two random geometric shapes A and B, turning function of a geometric shape A, turning function of a geometric shape B, and normalized arc length, respectively. Based on the experimental results, as the effective dissimilarity increases, the thermal performance and the operating range of PHPs are increased regardless of the input powers.

Under the excessively localized heating condition, it is almost impossible to design the channel layout using the effective dissimilarity. Hence, Lim and Kim [66] have developed a novel channel layout of a PHP using the topology optimization method as shown in Fig. 7(c). The topology optimization method is a relatively new approach in the field of heat transfer and distinguishes itself from conventional design methods in that this method does not require predefined configurations [67]. To account for the fundamental characteristics of PHPs, the constraints on the maximum length scale and the minimum solid volume fraction in local regions are imposed. It is experimentally shown that the PHP with the novel channel layout has a much higher thermal performance than the PHP with the uniform channel layout (Fig. 7(a)) regardless of the working fluids.

3.4 Optimization of the Channel Diameter and the Number of Turns. In practical applications, the channel design of PHPs is somewhat limited due to the constraint of the fixed space allowed for thermal management. Among the various design variables for the channel of PHPs, the channel diameter (D_h) and the number of turns (N) are key factors to be determined for optimizing the thermal performance of PHPs. The thermal performance of PHPs is enhanced with an increase of either the channel diameter [20,30,35] or the number of turns [33]. However, under the constraint of a fixed space, an increase of the channel diameter leads to a decrease in the number of turns, and vice versa. Correspondingly, an optimum channel diameter or an optimum number of turns must exist. Hence, Noh and Kim have performed a

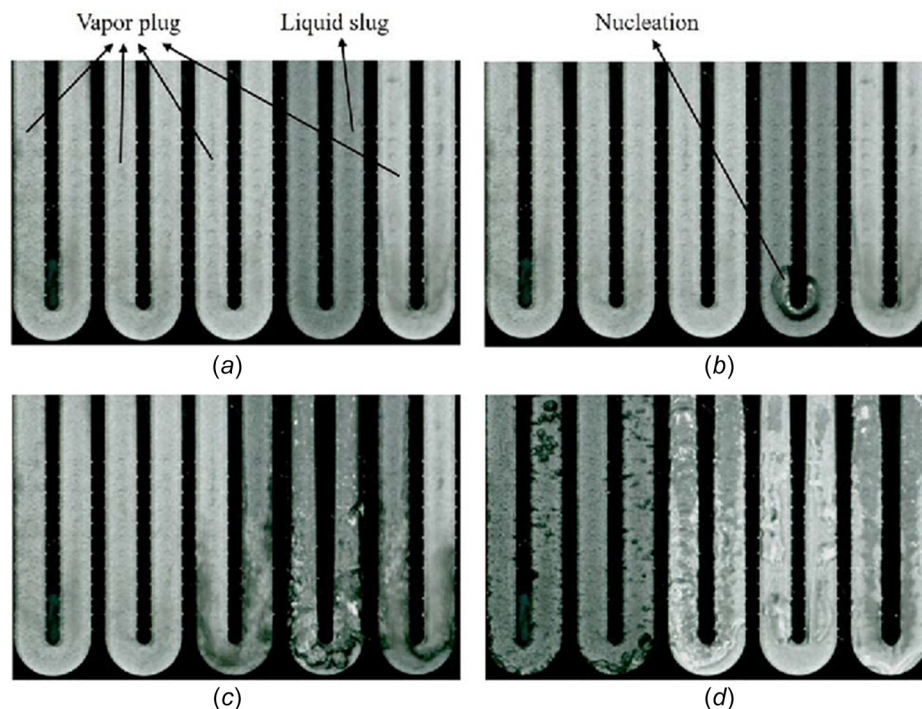


Fig. 6 Process of startup in the evaporator section of PHP with artificial cavities: (a) 0 s, (b) 0.5 ms, (c) 3.75 ms, and (d) 16.75 ms

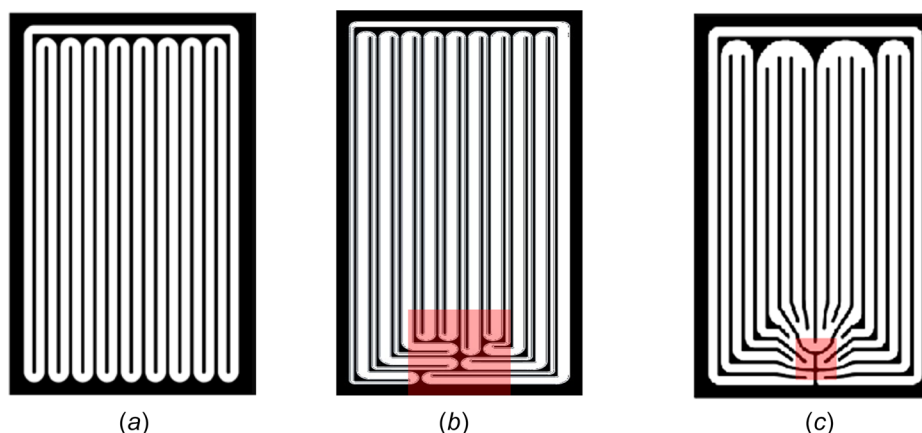


Fig. 7 PHPs with (a) uniform channel layout, (b) the developed channel layout using the effective dissimilarity, and (c) a novel channel layout using the topology optimization

numerical simulation to determine an optimum channel diameter or an optimum number of turns for PHPs. Based on their numerical results, it is found that $D_h^2 N$ can be used as a merit number for designing a PHP when other design variables excluding the channel diameter and the number of turns are fixed a priori. As shown in Fig. 8, the thermal performance of PHPs is maximized when the merit number is maximized.

3.5 Comparison Between Closed-Loop and Closed-End Pulsating Heat Pipes. A PHP can be classified into two types depending on its channel configuration [3,7,8]: a closed-loop PHP (CLPHP) and a closed-end PHP (CEPHP). To determine which type of a PHP performs better between a CLPHP and a CEPHP, many investigators have compared the thermal performance of a CLPHP and a CEPHP [11,24,33,68,69]. Yang et al. [69] have claimed that the CLPHP performs better than the CLPHP due to

possibility of a circulating flow inside a CLPHP, which is simply based on their conjecture. On the other hand, Jun and Kim [24] have reported that the CEPHP performs better than the CLPHP. In their results, a circulating flow was not visually observed inside CLPHPs.

Kim and Kim [26] have figured out the reason why there exist contradictory results by reproducing these results experimentally under different conditions. Based on their experimental results, a criterion of the diameter ratio ($D_h/D_{h,crit}$) is introduced to determine which type of a PHP performs better between the two types of PHPs: A CEPHP performs better than a CLPHP when $D_h/D_{h,crit}$ is lower than 0.5, while a CLPHP performs better than a CEPHP when $D_h/D_{h,crit}$ is higher than 0.5. When $D_h/D_{h,crit}$ exceeds 0.5, the crossover point in performance appears because the flow behavior inside the CLPHP changes from an oscillation mode to a circulation mode with the increasing thermal performance, while the flow behavior inside the CEPHP remains in an

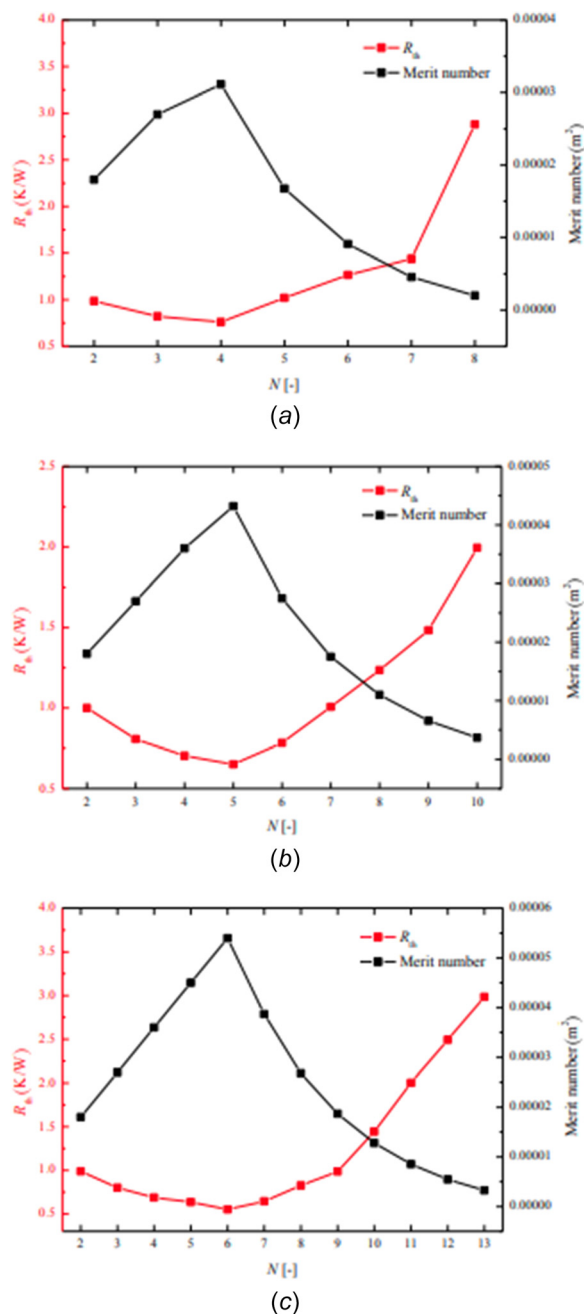


Fig. 8 Thermal resistances of PHPs and the merit numbers with a variation of the number of turns under conditions of fixed spaces with three widths ((a) 0.03, (b) 0.04, and (c) 0.05 m)

oscillation mode with negligible difference in thermal performance. These findings are consistent with the flow visualization data of the PHPs in the previous literature [20–30,69].

3.6 Selection of the Working Fluid. Depending on the operating temperature of PHPs, nitrogen, neon, hydrogen, and helium [70,71] can be selected as working fluids for cryogenic PHPs, while sodium [72] can be selected as the working fluid for high-temperature PHPs. In the case of room temperature applications, many possible candidates for working fluids have been investigated. To find the best working fluid for developing PHPs with high thermal performance, numerous researchers have experimentally compared the thermal performance of PHPs filled with various working fluids [73–75]. Hao et al. [73] have reported that the PHP filled with acetone shows the highest thermal performance

among PHPs filled with water, methanol, ethanol and acetone. Qu et al. [32] have compared the thermal performance of PHPs filled with water, ethanol, FC-72, and R-113, and experimentally demonstrated that R-113 is the best working fluid among them. Because it is impossible to compare the thermal performance of PHPs experimentally for all working fluids at a variety of operating temperatures, Kim and Kim [29] have provided a figure of merit, which can be used to select a working fluid for the maximum thermal performance of PHPs. The figure of merit for working fluid selection in PHPs is as follows:

$$M_{\text{PHP}} = \frac{\rho_l C_{p,l} (dP/dT)_{\text{sat}} Z R T_{\text{sat}}}{\mu_l h_{lv}} \quad (4)$$

where ρ_l , $C_{p,l}$, $(dP/dT)_{\text{sat}}$, Z , R , T_{sat} , μ_l , and h_{lv} are the liquid density, specific heat capacity, derivative of the vapor pressure with respect to the vapor temperature, compressibility factor, specific gas constant, and saturation temperature, liquid viscosity, and latent heat of vaporization, respectively. Figure 9 shows the variation of the figure of merit with respect to the operating temperature and the working fluid. The best working fluid for PHPs is the one with the highest value of M_{PHP} among the various working fluids. The figure of merit is consistent with the experimental results of previous studies [32,73–75].

3.7 Fabrication of Flexible Pulsating Heat Pipes. Even though many PHPs are fabricated using rigid materials such as copper, aluminum, glass and silicon, these rigid materials alone are not suitable for developing a flexible PHP. As a consequence, polymer has received much attention as an appropriate base material for a flexible PHP due to its high flexibility [76–79]. Nevertheless, long-term reliability is a major issue to be overcome for developing polymer-based PHPs due to the permeation of NCGs and the leakage of the working fluid [80–83]. Ogata et al. [78,79] fabricated a PHP using UV-curable polymer resin and polyethylene terephthalate film and observed the change in the thermal performance of the PHP over time. They glued a gas barrier film onto the top and bottom surfaces of the PHP to reduce the permeation and leakage, but their PHP was able to maintain its thermal performance only for 18 days, though [78,79]. Recently, Lim and Kim [84] and Jung et al. [85] have developed two generations of polymer-based flexible pulsating heat pipes (FPHPs), which significantly enhance the long-term reliability compared to the previous polymer-based PHP. Figure 10(a) shows the first-generation FPHP. It is experimentally confirmed that the lifetime of the first-generation FPHP is 306 days in standard atmosphere, which is 17 times longer than the previous polymer-based PHP [78,79]. The first-generation FPHP consists of a multilayer laminated film and a low-density polyethylene sheet. The heat-sealed flanges are

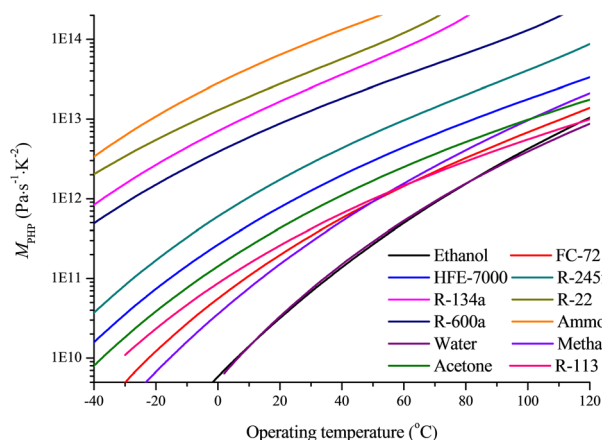


Fig. 9 Change in the values of the figure of merit with operating temperature and working fluid

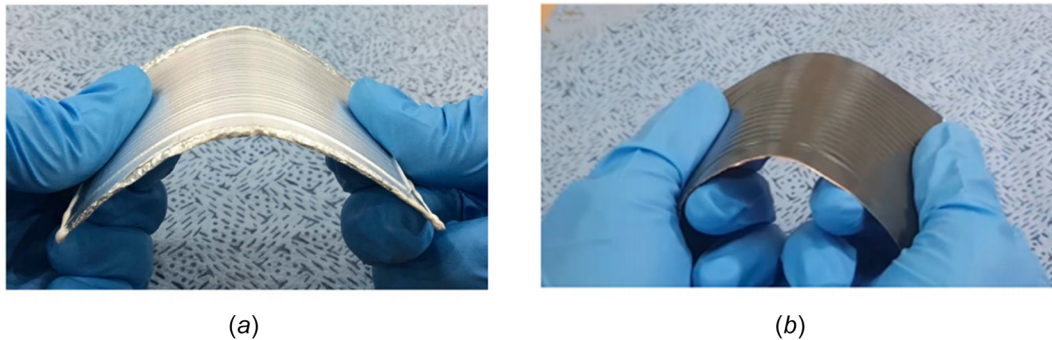


Fig. 10 Photographs of (a) the first-generation and (b) the second-generation FPHP

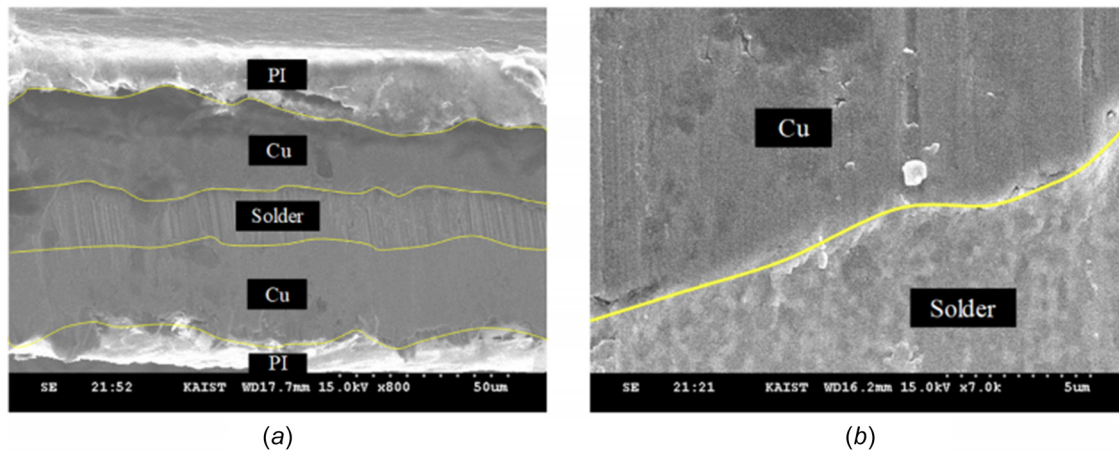


Fig. 11 (a) 800 times and (b) 7000 times magnification SEM images of the cross section of the flange joint of the second-generation FPHP

covered with an indium coating. To enhance not only the long-term reliability but also the thermal performance of PHPs, Jung et al. [85] have developed a second-generation FPHP as shown in Fig. 10(b). The lifetime of the second-generation FPHP is estimated to be at least 2000 days. The second-generation FPHP consists of a polycarbonate sheet and flexible-copper-clad-laminates (FCCLs). To improve the long-term reliability, the upper and lower sides of both channels and channel walls are encapsulated with the copper side of the FCCLs by a silane-mediated bonding method, and the flanges of the FPHP are sealed by soldering the two copper-foils as shown in Fig. 11. Moreover, the second-generation FPHP has 2.7 times higher thermal conductivity than copper.

4 Conclusion

In this paper, recent advances in PHPs are critically reviewed to put the finishing touches to the controversial issues on fundamental characteristics and to shed light on technical problems in practical applications. To this end, two categories are reviewed and highlighted: (i) fundamental aspects of PHPs and (ii) practical aspects of PHPs. The experimental/theoretical/numerical results of critical importance in each category are summarized as follows:

4.1 Fundamental Aspects of Pulsating Heat Pipes

4.1.1 Operating Mechanisms of PHPs. Using IR thermometry, it is experimentally confirmed that heat transfer in the form of latent heat is a dominant heat transfer mechanism. Moreover, it is concluded that the thermal performance of PHPs can be enhanced by increasing the proportion of latent heat to overall heat transferred through PHPs. Based on the flow visualization results, it is also found that an average volumetric fraction in the condenser

section is a key parameter governing the thermal performance of PHPs by affecting the flow behavior inside PHPs.

4.1.2 Unstable Operation of PHPs. It is found that dryout or stopover occurs in PHPs when the following conditions are met:

- Dryout occurs when the rate of liquid supply is smaller than the evaporation rate.
- If a PHP is treated as a mass-spring-damper system, stopover occurs when an effective damping coefficient becomes positive. For PHPs filled with ethanol, the effective damping coefficient reduces to D_h/L_{eff} and it has to be greater than approximately 0.03 for stable operation without any stopovers.

4.1.3 Numerical Models for PHPs. The earliest numerical models assumed the thickness of the liquid film does not vary with space and time. For predicting the thermal performance of PHPs without any fitting parameters even in a horizontal orientation, recent numerical models have been improved by allowing the thickness of the liquid film to be varied temporally and spatially. The most up-to-date numerical models are able to reasonably predict the thermal performance of PHPs, but they have shown very limited success in predicting dryout in PHPs, though. In addition, to help reduce the errors in the numerical models by implementing the proper thermodynamic state of vapor plugs, experiments have been conducted to confirm that a vapor plug in the PHP is saturated (superheated) if the vapor plug is (is not) surrounded by the liquid film.

4.2 Practical Aspects of Pulsating Heat Pipes

4.2.1 Omnidirectional Performance. Unless PHPs have a large number of turns (typically more than 20 turns), flow

restricting structures, including dual-diameter channels or check valves, need to be employed to achieve the omnidirectional performance even with only a few turns.

4.2.2 Extension of the Operating Range. It is experimentally confirmed that using either nanofluids or artificial cavities is helpful to promoting nucleation, which in turn leads to early startup. As a consequence, the operating range of PHPs can be extended using either nanofluids or artificial cavities.

4.2.3 Operation Under the Local Heating Condition. To prevent thermally isolated channels under the local heating condition, an effective dissimilarity is introduced to design the channel layout of PHPs. Additionally, to design the channel layout of PHPs under an excessively localized heating condition, the topology optimization method can be employed to develop a novel channel layout of a PHP.

4.2.4 Optimization of the Channel Diameter and the Number of Turns. Either an increase of the channel diameter (D_h) or the number of turns (N) increases the thermal performance of PHPs. However, under the constraint of a fixed space, an increase of the former leads to a decrease of the latter, and vice versa. Based on the numerical simulation, it is shown that $D_h^2 N$ should be maximized to optimize the thermal performance of PHPs.

4.2.5 Comparison Between Closed-Loop and Closed-End PHPs. A criterion of diameter ratio ($D_h/D_{h,crit}$) is introduced to determine which type of a PHP performs better between a CLPHP and a CEPHP: A CEPHP performs better than a CLPHP when $D_h/D_{h,crit}$ is lower than 0.5, while a CLPHP performs better than a CEPHP when $D_h/D_{h,crit}$ is higher than 0.5.

4.2.6 Selection of the Working Fluid. The figure of merit (M_{PHP}) is introduced as a guideline for selecting the best working fluid for PHPs. It is experimentally validated that the best working fluid for a PHP has the highest value of M_{PHP} among the various working fluids.

4.2.7 Fabrication of Flexible PHPs. A high-performance flexible PHP consists of a polycarbonate sheet, and flexible-copper-clad-laminates has been developed, and its lifetime has been estimated to be at least 2000 days. Moreover, the developed flexible PHP has 2.7 times higher thermal performance than copper.

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Nomenclature

- A = area (m^2)
 C_p = specific heat capacity ($J/kg \cdot K$)
 d = dissimilarity
 D_h = hydraulic diameter (m)
 h_{lv} = latent heat of vaporization (J/kg)
 L = length (m)
 N = number of turns
 r = normalized arc length
 T = temperature ($^{\circ}C$)

Greek Symbols

- $\bar{\beta}$ = average volumetric fraction
 ρ = density (kg/m^3)
 θ = turning function

Subscripts

- c = condenser
crit = critical

- eff = effective
l = liquid phase
omn = omnidirectional
PHP = pulsating heat pipe
sat = saturation
t = total
v = vapor phase
vp = vapor plug
 1ϕ = single-phase
 2ϕ = two-phase

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